

COURSE CODE/TITLE:	CPE212 Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	07/29/26	DATE SUBMITTED:	07/30/26
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ACTIVITY NO. 2.1	2.1 Hands-on Activity - SSH Key-Based Authentication and GIT Setup
1. PROCEDURE OUTPUT	

What procedures did you perform?

Task 1: Create an SSH Key Pair for User Authentication

1. The simplest way to generate a key pair is to run ssh-keygen without arguments. In this case, it will prompt for the file in which to store keys. First, the tool asked where to save the file. SSH keys for user authentication are usually stored in the users .ssh directory under the home directory. However, in enterprise environments, the location is often different. The default key file name depends on the algorithm, in this case id_rsa when using the default RSA algorithm. It could also be, for example, id_dsa or id_ecdsa.

2. Issue the command ssh-keygen -t rsa -b 4096. The algorithm is selected using the -t option and key size using the -b option.

3. When asked for a passphrase, just press enter. The passphrase is used for encrypting the key, so that it cannot be used even if someone obtains the private key file. The passphrase should be cryptographically strong.

4. Verify that you have created the key by issuing the command ls -la .ssh. The command should show the .ssh directory containing a pair of keys. For example, id_rsa.pub and id_rsa.

```

jerald@jerald:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/jerald/.ssh/id_rsa):
/home/jerald/.ssh/id_rsa already exists.
Overwrite (y/n)?
jerald@jerald:~$ ls -la ~/.ssh
total 28
drwx----- 2 jerald jerald 4096 Jul 29 22:38 .
drwxr-x--- 19 jerald jerald 4096 Jul 29 00:14 ..
-rw----- 1 jerald jerald  739 Jul 28 23:34 authorized_keys
-rw----- 1 jerald jerald 3381 Jul 28 23:29 id_rsa
-rw-r--r-- 1 jerald jerald  739 Jul 28 23:29 id_rsa.pub
-rw----- 1 jerald jerald 3218 Jul 29 22:49 known_hosts
-rw----- 1 jerald jerald 2098 Jul 29 22:37 known_hosts.old

```

Task 2: Copying the Public Key to the remote servers

1. To use public key authentication, the public key must be copied to a server and installed in an authorized_keys file. This can be conveniently done using the ssh-copy-id tool.
2. Issue the command similar to this: `ssh-copy-id -i ~/.ssh/id_rsa user@host`
3. Once the public key has been configured on the server, the server will allow any connecting user that has the private key to log in. During the login process, the client proves possession of the private key by digitally signing the key exchange.

Server1

```
jerald@jerald:~$ ssh-copy-id -i ~/.ssh/id_rsa interno@server1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/jerald/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
interno@server1's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'interno@server1'"
and check to make sure that only the key(s) you wanted were added.
```

Server2

```
jerald@jerald:~$ ssh-copy-id -i ~/.ssh/id_rsa internos2@server2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/jerald/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
internos2@server2's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'internos2@server2'"
and check to make sure that only the key(s) you wanted were added.
```

4. On the local machine, verify that you can SSH with Server 1 and Server 2.

Server1

```
jerald@jerald:~$ ssh interno@server1
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-28-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

*** System restart required ***
Last login: Wed Jul 29 22:43:14 2026 from 192.168.56.102
interno@server1:~$ exit
logout
Connection to server1 closed.
```

Server2

```
jerald@jerald:~$ ssh internos2@server2
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-28-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

*** System restart required ***
Last login: Wed Jul 29 14:43:26 2026 from 192.168.56.102
internos2@server2:~$ exit
logout
Connection to server2 closed.
```

What did you notice? Did the connection ask for a password? If not, why?

- The SSH session logs you directly into the remote server's terminal shell without asking for a account password.

Reflections:

Answer the following:

1. How will you describe the ssh-program? What does it do?
 - The ssh program is a secure networking tool that allows access and management of remote servers over an encrypted connection.
2. How do you know that you already installed the public key to the remote servers?
 - If the public key is installed, it can log in without entering a password, or if the public key's text appears inside the remote server's ~/.ssh/authorized_keys file.

Task 3: Set up the Git Repository

1. On the local machine, verify the version of your git using the command which git. If a directory of git is displayed, then you don't need to install git. Otherwise, to install git, use the following command: `sudo apt install git`

```
jerald@jerald:~$ which git
```

2. After the installation, issue the command which git again. The directory of git is usually installed in this location: `user/bin/git`.

```
internos2@server2: ~
jerald@jerald:~$ sudo apt update && sudo apt install git -y
Hit:1 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libwoff1 linux-headers-6.17.0-19-generic
  linux-hwe-6.17-headers-6.17.0-19 linux-hwe-6.17-tools-6.17.0-19
  linux-image-6.17.0-19-generic linux-modules-6.17.0-19-generic
  linux-modules-extra-6.17.0-19-generic linux-tools-6.17.0-19-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  git-man liberror-perl
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb
  git-cvs git-mediawiki git-svn
The following NEW packages will be installed:
  git git-man liberror-perl
```

```
jerald@jerald:~$ which git
/usr/bin/git
```

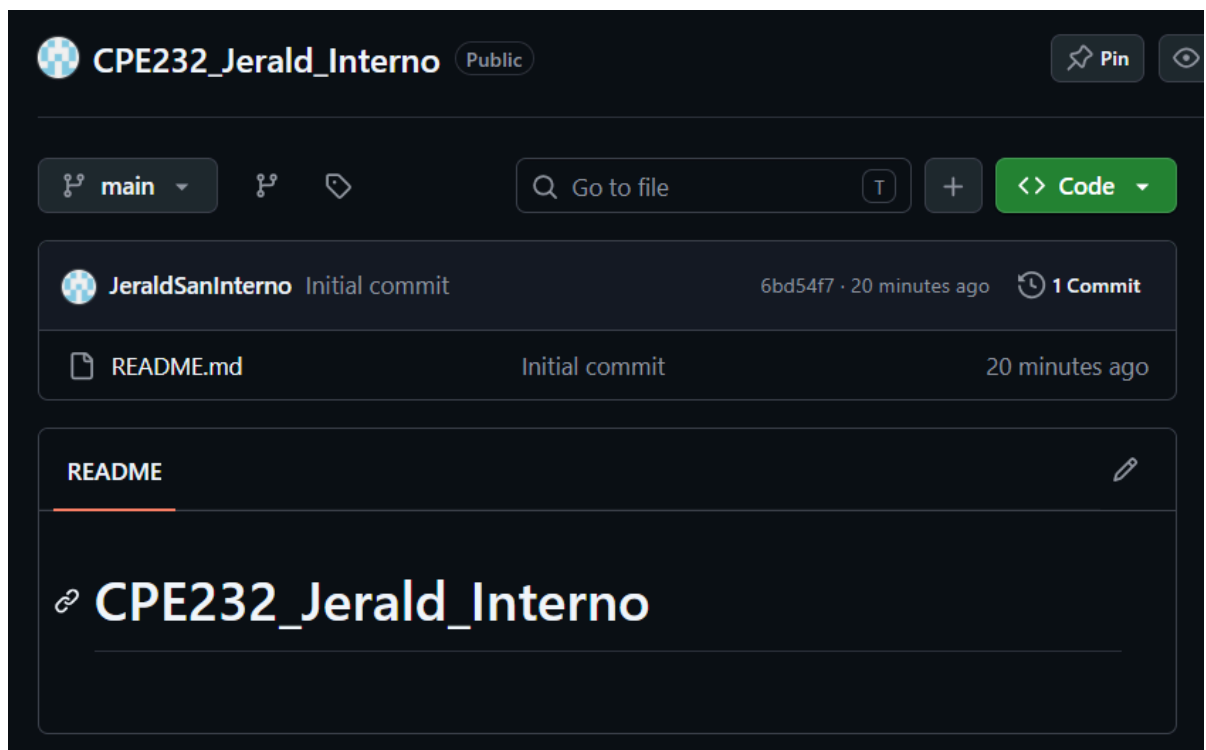
3. The version of git installed in your device is the latest. Try issuing the command `git --version` to know the version installed.

```
jerald@jerald:~$ git --version
git version 2.43.0
```

4. Using the browser in the local machine, go to www.github.com.

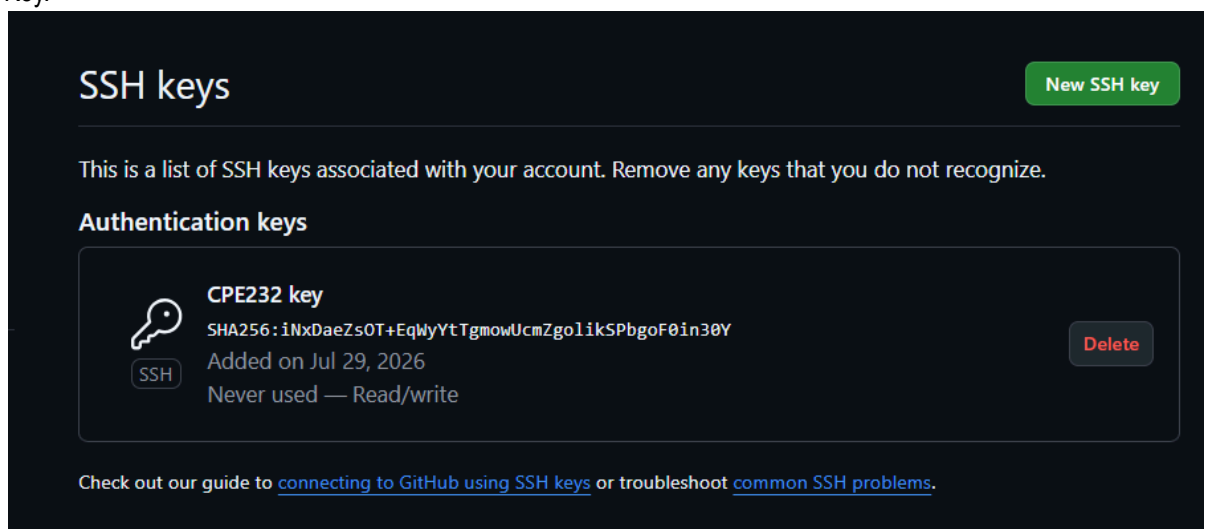
5. Sign up in case you don't have an account yet. Otherwise, login to your GitHub Account.

a. Create a new repository and name it as `CPE232_yourname`. Check Add a README file and click Create repository.



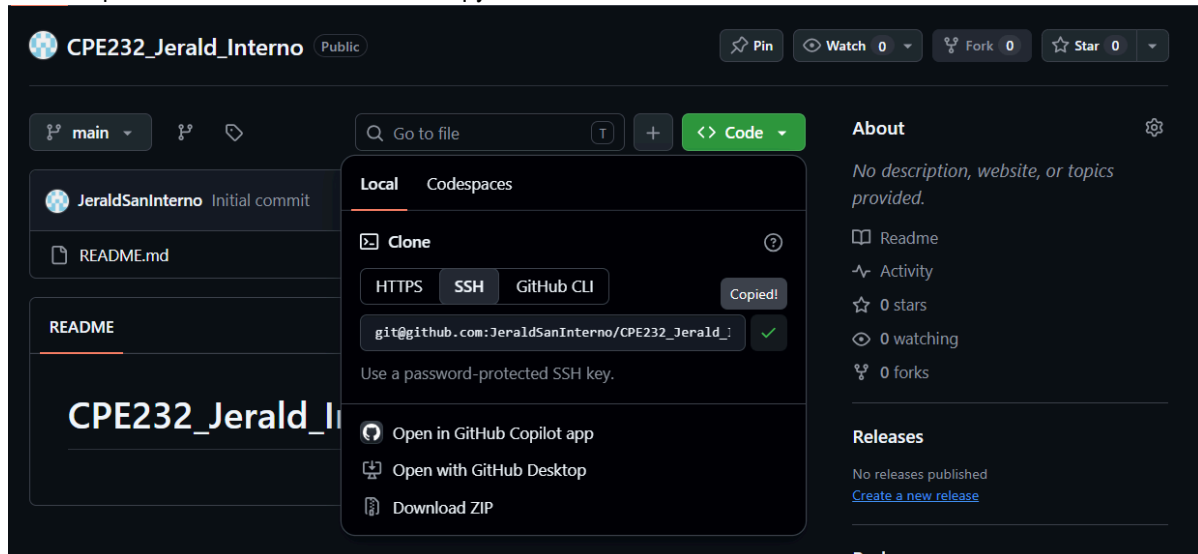
b. Create a new SSH key on GitHub. Go your profile's setting and click SSH and GPG keys. If there is an existing key, make sure to delete it. To create a new SSH keys, click New SSH Key. Write CPE232 key as the title of the key.

c. On the local machine's terminal, issue the command `cat .ssh/id_rsa.pub` and copy the public key. Paste it on the GitHub key and press Add SSH Key.



d. Clone the repository that you created. In doing this, you need to get the link from GitHub. Browse to your repository as shown below. Click on the

Code drop down menu. Select SSH and copy the link.



e. Issue the command `git clone` followed by the copied link. For example, `git clone git@github.com:jvtaylor-cpe/CPE232_yourname.git`. When prompted to continue connecting, type `yes` and press enter.

```
jerald@jerald:~$ git clone git@github.com:JeraldSanInterno/CPE232_Jerald_Interno.git
Cloning into 'CPE232_Jerald_Interno'...
The authenticity of host 'github.com (20.205.243.166)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvV6TuJJhbpZisF/zLDA0zPMSvHdKr4UvCOqU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

f. To verify that you have cloned the GitHub repository, issue the command `ls`. Observe that you have the `CPE232_yourname` in the list of your directories. Use `CD` command to go to that directory and `LS` command to see the file [README.md](#).

```
jerald@jerald:~$ ls
ansible-demo      Desktop           Documents         Music             Public            sort_files.sh     trial
CPE232_Jerald_Interno  Directories      Downloads         Pictures          snap              Templates          Videos
jerald@jerald:~$ cd CPE232_Jerald_Interno
jerald@jerald:~/CPE232_Jerald_Interno$ ls -la
.  .. .git README.md
```

g. Use the following commands to personalize your git.

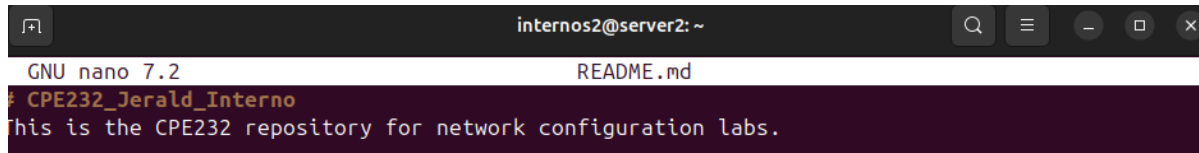
- `git config --global user.name "Your Name"`
- `git config --global user.email yourname@email.com`

```
jerald@jerald:~/CPE232_Jerald_Interno$ git config --global user.name Jerald
jerald@jerald:~/CPE232_Jerald_Interno$ git config --global user.email qjsminterno@tip.edu.ph
```

• Verify that you have personalized the config file using the command `cat ~/.gitconfig`

```
jerald@jerald:~/CPE232_Jerald_Interno$ git config --global user.name Jerald
jerald@jerald:~/CPE232_Jerald_Interno$ git config --global user.email qjsminterno@tip.edu.ph
jerald@jerald:~/CPE232_Jerald_Interno$ cat ~/.gitconfig
[user]
    name = Jerald
    email = qjsminterno@tip.edu.ph
```

h. Edit the README.md file using nano command. Provide any information on the markdown file pertaining to the repository you created. Make sure to write out or save the file and exit.



```
internos2@server2: ~
GNU nano 7.2                                README.md
# CPE232_Jerald_Interno
This is the CPE232 repository for network configuration labs.
```

i. Use the git status command to display the state of the working directory and the staging area. This command shows which changes have been staged, which haven't, and which files aren't being tracked by Git. Status output does not show any information regarding the committed project history.

```
jerald@jerald:~/CPE232_Jerald_Interno$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
```

What is the result of issuing this command?

- **The terminal output shows Changes not staged for commit with modified: README.md highlighted in red. This indicates Git detected local edits in README.md that haven't been staged yet.**

j. Use the command git add README.md to add the file into the staging area.

```
jerald@jerald:~/CPE232_Jerald_Interno$ git add README.md
```

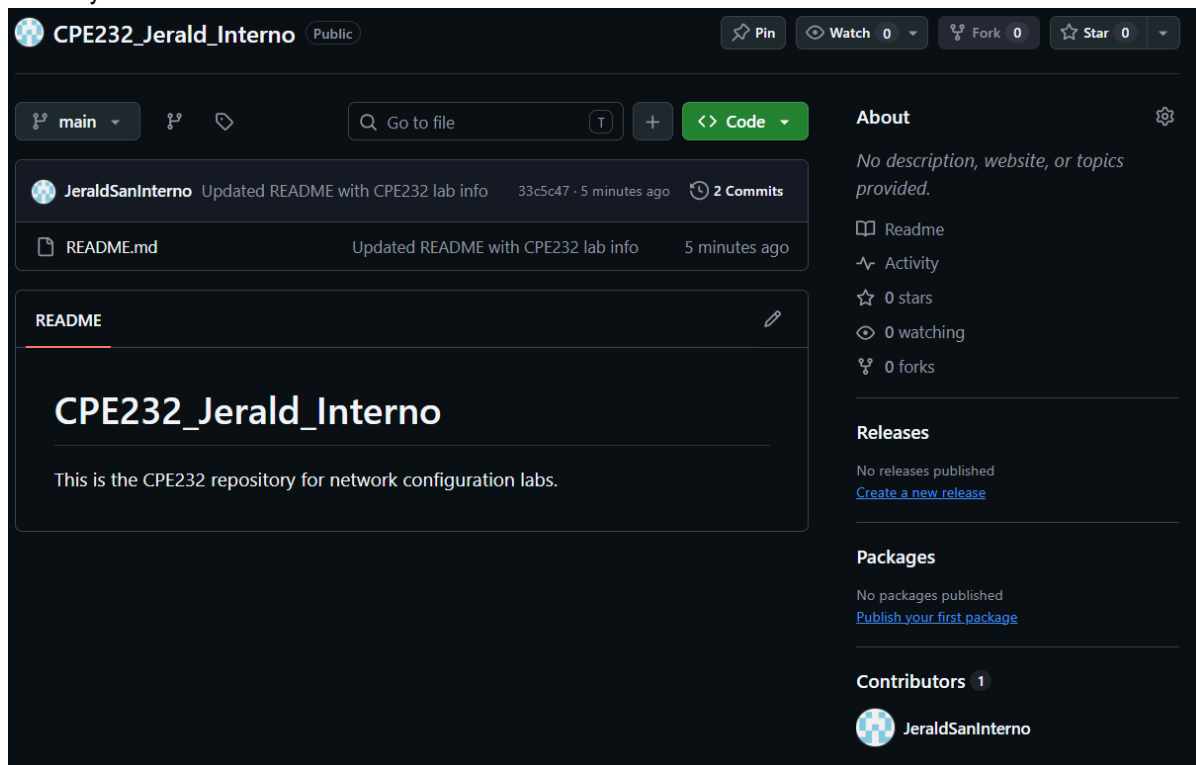
k. Use the git commit -m "your message" to create a snapshot of the staged changes along the timeline of the Git projects history. The use of this command is required to select the changes that will be staged for the next commit.

```
jerald@jerald:~/CPE232_Jerald_Interno$ git commit -m "Updated README with CPE232 lab info"
[main 33c5c47] Updated README with CPE232 lab info
1 file changed, 2 insertions(+), 1 deletion(-)
```

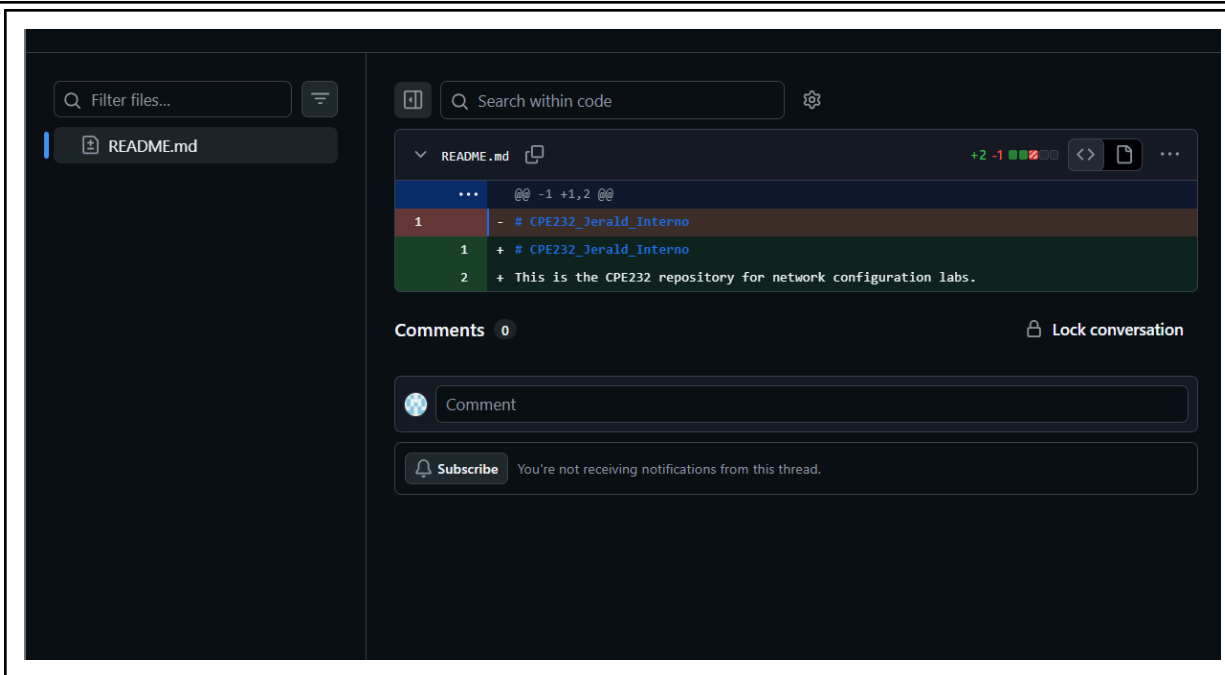
l. Use the command git push <remote><branch> to upload the local repository content to GitHub repository. Pushing means to transfer commits from the local repository to the remote repository. As an example, you may issue git push origin main.

```
jerald@jerald:~/CPE232_Jerald_Interno$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 6 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 347 bytes | 347.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:JeraldSanInterno/CPE232_Jerald_Interno.git
6bd54f7..33c5c47  main -> main
```

m. On the GitHub repository, verify that the changes have been made to README.md by refreshing the page. Describe the README.md file. You can notice the how long was the last commit. It should be some minutes ago and the message you typed on the git commit command should be there. Also, the README.md file should have been edited according to the text you wrote.



The screenshot shows the GitHub repository page for **CPE232_Jerald_Interno**, which is a public repository. The repository is owned by **JeraldSanInterno**. The main branch is selected. The repository has 0 watches, 0 forks, and 0 stars. The README file is highlighted, showing the title **CPE232_Jerald_Interno** and the description: "This is the CPE232 repository for network configuration labs." The commit history shows a recent commit by JeraldSanInterno titled "Updated README with CPE232 lab info" 5 minutes ago, with 2 commits in total. The right sidebar shows the repository's metadata, including the README, activity, stars, watching, forks, releases, packages, and contributors.



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Reflections:

Answer the following:

3. What sort of things have we so far done to the remote servers using ansible commands?

- Using Ansible commands, I have connected to remote servers via SSH, gathered system information, pinged hosts to verify connectivity, and executed remote commands or configuration changes.

4. How important is the inventory file?

- The inventory file is crucial since it defines and organizes the list of target hosts (remote servers) and their IP addresses or domains that Ansible manages.

Conclusions/Learnings:

- In this lab, we learned how to set up secure, passwordless authentication using SSH keys for both remote servers and GitHub. We practiced the core Git workflow by cloning a repository, customizing our user configuration, editing files, and pushing our local changes to GitHub. Finally, we learned that Ansible relies on these secure SSH connections and an inventory file to easily manage and control remote servers from one central place.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Gemini

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.



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Purpose of Use: grammar correction, idea generation, explanation of concepts.

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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